

1. Yes, $x_1^* = 1, x_2^* = 2$ is a feasible solution. The verification is done by substitution:

$$\begin{array}{rcl} 1 + 2 = 3 & \geq & 3, \\ 1 + 2 \cdot 2 = 5 & \leq & 6, \\ 1 & \geq & 1, \\ 2 & \geq & 1. \end{array} \quad \begin{array}{c} \checkmark \\ \checkmark \\ \checkmark \\ \checkmark \end{array}$$

The first and the third constraints are active at (x_1^*, x_2^*) . They are the two constraints that are verified with equality. The point lies at the intersection of the two lines that have $x_1 + x_2 = 3$ and $x_1 = 1$ as equations and is labeled A in the figure below.

2. Yes, $x_1^* = 2, x_2^* = 1.5$ is a feasible solution. The verification is done by substitution:

$$\begin{array}{rcl} 2 + 1.5 = 3.5 & \geq & 3 \\ 2 + 2 \cdot 1.5 = 5 & \leq & 6 \\ 2 & \geq & 1 \\ 1.5 & \geq & 1. \end{array} \quad \begin{array}{c} \checkmark \\ \checkmark \\ \checkmark \\ \checkmark \end{array}$$

No constraint is active. Therefore, (x_1^*, x_2^*) is an interior point. It is labeled B in the figure below.

3. The point $x_1^* = 1, x_2^* = 1$ is infeasible, as it violates the first constraint.

$$\begin{array}{rcl} 1 + 1 = 2 & \not\geq & 3 \\ 1 + 2 \cdot 1 = 3 & \leq & 6 \\ 1 & \geq & 1 \\ 1 & \geq & 1. \end{array} \quad \begin{array}{c} \times \\ \checkmark \\ \checkmark \\ \checkmark \end{array}$$

The point is labeled C in the figure below.

The graph below shows the feasible region for the problem.

